3 A programmer is designing a computer game that uses a set of cards.

Each card has a number and a colour. The cards are saved in the text file `CardValues.txt`

The program has a class named `Card`. The class has the following attributes and methods.

Card	
`Number : INTEGER`	The number of the card
`Colour : STRING`	The colour of the card
`Constructor()`	Takes two values as parameters and sets them to the private attributes
`GetNumber()`	Returns the number of the card
`GetColour()`	Returns the colour of the card

(a) The constructor takes the number and colour of the card as parameters and sets them to the private attributes.

Write program code to declare the class `Card` and its constructor. Do **not** write any other methods.

Use your programming language appropriate constructor.

All attributes should be private. If you are writing in Python, include attribute declarations using comments.

Save your program as **Question3_J22**.

Copy and paste the program code into **part 3(a)** in the evidence document.

[5]

(b) The two get methods return the associated attribute.

Write program code for the get methods `GetNumber()` and `GetColour()`.

Save your program.

Copy and paste the program code into **part 3(b)** in the evidence document.

[3]

(c) The text file `CardValues.txt` stores the data for 30 cards, in the order: number, colour.

For example, the first card in the text file:

`1` is the number
`red` is the colour.

A 1D array of type `Card` is declared to store all the cards read in from `CardValues.txt`

Write the main program to:

- declare an array of type `Card` with 30 elements
- read in the data for the 30 cards from `CardValues.txt` and assign each to the array.

Save your program. Copy and paste the program code into **part 3(c)** in the evidence document.

[7]

(d) The program needs to allow all players (maximum of 5) to select 4 cards from the 30 available. A card can only be selected once, so the program needs to record which cards have already been selected.

The function, `ChooseCard()`:

- takes as input an integer to represent an array index from 1 to 30
- validates that the value is between 1 and 30 inclusive
- checks if the card is available (it has not already been selected)
- loops until an available card is selected
- returns the index of the card if it is available.

Amend the program to store which cards have already been selected **and** write program code for the function `ChooseCard()`.

Save your program. Copy and paste the program code into **part 3(d)** in the evidence document.

[6]

(e) The main program needs to allow one player to select all their 4 cards.

(i) Amend the main program to:

- create an array, `Player1`, for player 1 of type `Card`
- ask player 1 to input 4 integers using the function from **part 3(d)**
- store the cards in `Player1`
- output the number and colour of the 4 cards in `Player1`.

Save your program. Copy and paste the program code into **part 3(e)(i)** in the evidence document.

[5]

(ii) Test your program with the following test data:

Test 1: **1** **5** **9** **10**

Test 2: **2** **2** **3** **4** **4** **5**

Take a screenshot to show the output.

Copy and paste the screenshot into **part 3(e)(ii)** in the evidence document.

[1]